

Mathew Fraser Robertson

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Data Scientist with a PhD in Physics and multiple first-author scientific publications; I developed new statistical methods to detect subtle trends from complex datasets. I have industry experience deploying agile machine learning software for banks in a client-facing role at Featurespace, and a significant quantitative data analysis background across the UK, Germany, and Saudi Arabia.

Education

University of Sussex

Sept 2021 – Jul 2025

Physics PhD (Awarded)

Brighton, UK

Key Publications: “How to detect lensing rotation” ([JCAP08\(2023\)048](#)), “Detectable signals of post-Born lensing curl B-modes” ([JCAP02\(2025\)034](#)).

- Lead investigator in several collaborative research projects resulting in high-impact first-author publications and multiple presentations at international conferences.
- Improved lensing sensitivity by 3 orders of magnitude through the invention of new statistical methods, enabling detections within 2 years instead of 20.
- Optimised Bayesian forecasting and numerical simulation codes by a factor of 1000s through vectorisations and parallel programming.
- Research also involved analytical modelling, high-performance computing (HPC), data visualisation, and mentoring of junior PhD and master’s students.
- Over 2 years as sole organiser for the Sussex cosmology research seminars.
- Key technologies: Python (Matplotlib, NumPy, SciPy, Pandas, Jupyter, mpi4py), Linux, HPC, Bash, Git, GitHub, and N-body Simulations, Gaussian Simulations, and Data Visualisation.

University of Glasgow

Sept 2014 – Jun 2019

Physics Masters (1st Class Honours)

Glasgow, UK

- **Courses Include:** Problem Solving Physics, Astronomical Data Analysis, Numerical Methods.
- **Master’s Project:** Analysed dataset of over 2 million galaxies to optimise target selection of telescope pointings following gravitational wave detections.
- Key technologies: Python (Matplotlib, NumPy, SciPy, Pandas, Jupyter), Linux, HPC, Bash, Cython, Big Data, and Data Visualisation.

University of Massachusetts at Amherst

Sept 2015 – Jun 2016

Study Abroad (3.6 GPA; 2016 Dean’s List)

Amherst, MA, USA

- **Courses Include:** Computational Physics, Calculus 1 & 2, Linear Algebra 1 & 2.

Work Experience

Featurespace

Nov 2020 – Aug 2021

Implementation Engineer

Cambridge, UK

- Lead engineer on 3 client-facing projects: I oversaw the design, implementation and testing (CI/CD), deployment, and monitoring of bespoke machine learning software for banks.
- Key technologies: Python, NoSQL (MongoDB), Git (GitLab), Linux, REST APIs, CI/CD, monitoring dashboards, and exposure to cloud (AWS).

KAUST

May 2019 – Oct 2019

Computational Materials Science Research Intern

Jeddah, Saudi Arabia

- Improved efficiency of solar panels designed with cheaper materials through advanced computational modelling and data analysis.
- Developed a scientific software package to extract statistics from simulated data accurately.
- Key technologies: Python (NumPy, SciPy), Linux, HPC, Bash, and QuantumEspresso.

DESY

Jun 2018 – Sept 2018

Computational Particle Physics Research Intern

Hamburg, Germany

- Optimised signal regions for new physics searches with the ATLAS detector at CERN through analysis of powerful Monte-Carlo simulations.
- Reduced modelling errors by 100% by creating a new algorithm to account for missing and duplicate data.
- Key technologies: C++, Linux, HPC, Bash, Root, GitLab, and Monte Carlo Simulations.

University of Glasgow

Jun 2017 – Sept 2017

Experimental Particle Physics Research Intern

Glasgow, UK

- Increased the accuracy of CERN simulations by 15% through tuning key parameters and implementing improvements to statistical software.
- Key technologies: Python (NumPy), Linux, HPC, Bash, and Monte Carlo Simulations.

University of Glasgow

Jun 2016 – Sept 2016

Statistics Research Intern

Glasgow, UK

- Reduced systematic errors in the calibration of Earth observation satellites through the development of advanced regression models.
- Key technologies: Python (Matplotlib, NumPy, Pandas), R, Bootstrap Simulations.

Decathlon

Apr 2014 – Oct 2014

Sports Adviser

Glasgow, UK

- Oversaw sales increase of 20% on aisle I managed through analysis of layout optimisation.
- Expert advisor on running; I analysed customers' gait in person to give bespoke advice.

Complementary Information

Awards

- STFC Postgraduate Research Scholarship (2021 - 2025)
- Sir Godfrey Collins Travelling Scholarship (2019)
- Engineering and Physical Sciences Research Council Vacation Scholarship (2016)
- Bronze British Science Award (2014)
- D G Smith Award (2014)

Teaching

- Doctoral tutor for Mathematical Methods & Physics in Finance at the University of Sussex (2023 - 2024).
- Physics tutor for Level 1 & 2 Physics at the University of Glasgow (2018 - 2019).

Interests

- Video game development (Unity, C#, JavaScript, HTML, Blender).
- Building desktop applications/GUIs (Python, SQL, Java, Qt)
- Solving mathematical coding challenges on Project Euler (Python).
- Machine learning (completed senior Stanford CS229 online course).
- As a sports enthusiast, I enjoy playing football, bouldering, and racquet sports in my free time.